

# THE PROBABILITY HEIST

TEACHER ANSWER KEY · TEKS 7.6 (A, D, H, I)

|   |   |   |   |   |   |   |   |   |    |    |    |    |    |    |
|---|---|---|---|---|---|---|---|---|----|----|----|----|----|----|
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 |
| C | R | A | C | K | E | D | T | H | E  | C  | A  | S  | E  | !  |

Final reveal phrase: **CRACKED THE CASE!**

## CROSSWALK · STUDENT CLUE LETTER → SLOT NUMBER

Students see clues labeled A–O in scrambled order. Each clue places its character into a numbered slot (1–15) on the Evidence Board. Use this table to grade quickly.

| Clue | Slot | TEKS   | Source   | Title                   | Answer     | Char |
|------|------|--------|----------|-------------------------|------------|------|
| A    | 13   | 7.6(I) | 2024 Q13 | The Twin Spinners       | 1/12 (12)  | S    |
| B    | 7    | 7.6(H) | 2024 Q6  | The Library Card Trail  | ACTION     | D    |
| C    | 2    | 7.6(D) | 2025 Q6  | The Carnival Spinner    | 15         | R    |
| D    | 14   | 7.6(I) | 2023 Q16 | The 8-Faced Solid       | 1/15 (15)  | E    |
| E    | 1    | 7.6(A) | 2023 Q2  | The Pizzeria Alibi      | 6          | C    |
| F    | 9    | 7.6(H) | 2023 Q15 | The Spinner of Shapes   | A          | H    |
| G    | 5    | 7.6(H) | 2025 Q1  | The Ice-Cream Parlor    | C          | K    |
| H    | 11   | 7.6(I) | 2025 Q13 | The Box of Keys         | 13/40 (13) | C    |
| I    | 3    | 7.6(D) | 2025 Q6  | The Carnival Spinner II | 75         | A    |
| J    | 15   | 7.6(I) | 2023 Q30 | The Deli Counter        | 1/12 (12)  | !    |
| K    | 8    | 7.6(H) | 2024 Q29 | The Pet Shop Witness    | D          | T    |
| L    | 4    | 7.6(D) | 2023 Q6  | The Marked Deck         | 6          | C    |
| M    | 12   | 7.6(I) | 2025 Q24 | The Compound Card Trick | 1/40 (40)  | A    |
| N    | 6    | 7.6(H) | 2025 Q19 | The Grocer's Ledger     | B          | E    |
| O    | 10   | 7.6(H) | 2023 Q35 | The Basketball Records  | LESS       | E    |

## ASSEMBLY VERIFICATION · SLOT ORDER (1 → 15)

Slots 1–15 in numerical order. Reading down the 'Char' column gives the final phrase: **CRACKED THE CASE!**

| Slot | Char | From Clue | TEKS   | Title                   |
|------|------|-----------|--------|-------------------------|
| 1    | C    | E         | 7.6(A) | The Pizzeria Alibi      |
| 2    | R    | C         | 7.6(D) | The Carnival Spinner    |
| 3    | A    | I         | 7.6(D) | The Carnival Spinner II |
| 4    | C    | L         | 7.6(D) | The Marked Deck         |

|    |          |   |        |                         |
|----|----------|---|--------|-------------------------|
| 5  | <b>K</b> | G | 7.6(H) | The Ice-Cream Parlor    |
| 6  | <b>E</b> | N | 7.6(H) | The Grocer's Ledger     |
| 7  | <b>D</b> | B | 7.6(H) | The Library Card Trail  |
| 8  | <b>T</b> | K | 7.6(H) | The Pet Shop Witness    |
| 9  | <b>H</b> | F | 7.6(H) | The Spinner of Shapes   |
| 10 | <b>E</b> | O | 7.6(H) | The Basketball Records  |
| 11 | <b>C</b> | H | 7.6(I) | The Box of Keys         |
| 12 | <b>A</b> | M | 7.6(I) | The Compound Card Trick |
| 13 | <b>S</b> | A | 7.6(I) | The Twin Spinners       |
| 14 | <b>E</b> | D | 7.6(I) | The 8-Faced Solid       |
| 15 | <b>!</b> | J | 7.6(I) | The Deli Counter        |

## DETAILED SOLUTIONS · BY STUDENT CLUE LETTER

|                   |                 |                                                    |             |
|-------------------|-----------------|----------------------------------------------------|-------------|
| <b>CLUE<br/>A</b> | <b>Slot #13</b> | <b>The Twin Spinners</b><br>TEKS 7.6(I) · 2024 Q13 | <b>1/12</b> |
|-------------------|-----------------|----------------------------------------------------|-------------|

**Solution:**  $P(X = 2) = 1/4$ .  $P(Y = \text{Blue}) = 1/3$ . Independent events:  $1/4 \times 1/3 = 1/12$ .

**Misconception watch:** Common error (C: 2/7) — students added denominators (4+3) instead of multiplying (4×3) for compound events.

|                   |                |                                                        |               |
|-------------------|----------------|--------------------------------------------------------|---------------|
| <b>CLUE<br/>B</b> | <b>Slot #7</b> | <b>The Library Card Trail</b><br>TEKS 7.6(H) · 2024 Q6 | <b>ACTION</b> |
|-------------------|----------------|--------------------------------------------------------|---------------|

**Solution:** Verify each option:  $2 \times \text{Nonfiction (9)} = 18$ . Action (24) > 18 ✓. So student is more than twice as likely to prefer Action as Nonfiction.

**Misconception watch:** Only 11% earned full credit on the original drag-and-drop; 51% partial. Students often skipped checking BOTH conditions in compound statements.

|                   |                |                                                      |           |
|-------------------|----------------|------------------------------------------------------|-----------|
| <b>CLUE<br/>C</b> | <b>Slot #2</b> | <b>The Carnival Spinner</b><br>TEKS 7.6(D) · 2025 Q6 | <b>15</b> |
|-------------------|----------------|------------------------------------------------------|-----------|

**Solution:**  $P(20) = 1/10$  (one section out of ten).  $150 \text{ spins} \times 1/10 = 15$ .

**Misconception watch:** Only 33% earned full credit on the original two-part item statewide. Reinforce: count favorable sections / total sections, then multiply by trials.

|                   |                 |                                                    |             |
|-------------------|-----------------|----------------------------------------------------|-------------|
| <b>CLUE<br/>D</b> | <b>Slot #14</b> | <b>The 8-Faced Solid</b><br>TEKS 7.6(I) · 2023 Q16 | <b>1/15</b> |
|-------------------|-----------------|----------------------------------------------------|-------------|

**Solution:** 7 appeared 2 times in 30 rolls.  $P(7) = 2/30 = 1/15$  simplified.

**Misconception watch:** Only 46% correct. Students confused experimental (data-based) with theoretical (1/8) probability. Watch for B (1/8) as the high-frequency wrong answer.

|                   |                |                                                    |          |
|-------------------|----------------|----------------------------------------------------|----------|
| <b>CLUE<br/>E</b> | <b>Slot #1</b> | <b>The Pizzeria Alibi</b><br>TEKS 7.6(A) · 2023 Q2 | <b>6</b> |
|-------------------|----------------|----------------------------------------------------|----------|

**Solution:** 2 toppings × 3 crusts = **6 total pizzas**. Sample space: (P,Thin), (P,Thick), (P,Stuffed), (S,Thin), (S,Thick), (S,Stuffed).

**Misconception watch:** 67% earned full credit on the original. Missed pairs are usually the result of unsystematic listing — push tree diagrams or organized tables.

|                   |                |                                                        |          |
|-------------------|----------------|--------------------------------------------------------|----------|
| <b>CLUE<br/>F</b> | <b>Slot #9</b> | <b>The Spinner of Shapes</b><br>TEKS 7.6(H) · 2023 Q15 | <b>A</b> |
|-------------------|----------------|--------------------------------------------------------|----------|

**Solution:** 4 diamonds > 3 stars > 1 triangle. So "more likely diamond than star" is TRUE. (B false: not equal; C false: 4 vs 3 ≠ 4×; D false: 4 > 3.)

**Misconception watch:** 72% correct on the original. C (15%) was the high-frequency wrong answer — students saw "4" diamonds and assumed "4 times as likely."

|                   |                |                                                      |          |
|-------------------|----------------|------------------------------------------------------|----------|
| <b>CLUE<br/>G</b> | <b>Slot #5</b> | <b>The Ice-Cream Parlor</b><br>TEKS 7.6(H) · 2025 Q1 | <b>C</b> |
|-------------------|----------------|------------------------------------------------------|----------|

**Solution:** Verify ratios: Chocolate (28) = 2 × Sherbet (14) ✓. This holds for any sample size including the next 350.

**Misconception watch:** B (Vanilla = 2× Sherbet) was the highly-chosen wrong answer. Students did not verify:  $16 \neq 2 \times 14$ .

|                   |                 |                                                  |              |
|-------------------|-----------------|--------------------------------------------------|--------------|
| <b>CLUE<br/>H</b> | <b>Slot #11</b> | <b>The Box of Keys</b><br>TEKS 7.6(I) · 2025 Q13 | <b>13/40</b> |
|-------------------|-----------------|--------------------------------------------------|--------------|

**Solution:** Yellow (5) + Green (8) = 13 favorable out of 40 trials.  $P(\text{yellow OR green}) = 13/40$ .

**Misconception watch:** 60% correct. C (2/5) was the high-frequency wrong answer — students used theoretical reasoning (2 of 5 colors) instead of experimental frequencies.

|               |                |                                                            |           |
|---------------|----------------|------------------------------------------------------------|-----------|
| <b>CLUE I</b> | <b>Slot #3</b> | <b>The Carnival Spinner Pt II</b><br>TEKS 7.6(D) · 2025 Q6 | <b>75</b> |
|---------------|----------------|------------------------------------------------------------|-----------|

**Solution:** 5 of 10 sections labeled 0 or 5.  $P = 5/10 = 1/2$ .  $150 \times 1/2 = 75$ .

**Misconception watch:** Same source item as Clue C. On the original, students often missed the second blank because they did not re-count favorable sections for the new event.

|               |                 |                                                   |             |
|---------------|-----------------|---------------------------------------------------|-------------|
| <b>CLUE J</b> | <b>Slot #15</b> | <b>The Deli Counter</b><br>TEKS 7.6(I) · 2023 Q30 | <b>1/12</b> |
|---------------|-----------------|---------------------------------------------------|-------------|

**Solution:** Sample space: 2 breads  $\times$  3 meats  $\times$  2 cheeses = 12 sandwiches. Only 1 matches White + Ham + Cheddar.  $P = 1/12$ .

**Misconception watch:** Only 26% correct statewide. 37% chose B (1/3) — students counted only one factor instead of multiplying across all three categories.

|               |                |                                                       |          |
|---------------|----------------|-------------------------------------------------------|----------|
| <b>CLUE K</b> | <b>Slot #8</b> | <b>The Pet Shop Witness</b><br>TEKS 7.6(H) · 2024 Q29 | <b>D</b> |
|---------------|----------------|-------------------------------------------------------|----------|

**Solution:** Dog (110) > Rabbit, Cat, Fish individually. So student is more likely to prefer dog than any other type.

**Misconception watch:** 71% correct on the original. B was high-frequency wrong — students saw "5 times" and matched fish:rabbit ratio without verifying  $15 \neq 5 \times 10$ .

|               |                |                                                 |          |
|---------------|----------------|-------------------------------------------------|----------|
| <b>CLUE L</b> | <b>Slot #4</b> | <b>The Marked Deck</b><br>TEKS 7.6(D) · 2023 Q6 | <b>6</b> |
|---------------|----------------|-------------------------------------------------|----------|

**Solution:** Total = 12 + 16 = 28.  $P(\text{letter}) = 12/28 = 3/7$ . 14 trials  $\times$   $3/7 = 6$ .

**Misconception watch:** 52% correct on the original. Many students set up  $14 \times 12/28$  but forgot to simplify or stopped at the fraction.

|               |                 |                                                          |             |
|---------------|-----------------|----------------------------------------------------------|-------------|
| <b>CLUE M</b> | <b>Slot #12</b> | <b>The Compound Card Trick</b><br>TEKS 7.6(I) · 2025 Q24 | <b>1/40</b> |
|---------------|-----------------|----------------------------------------------------------|-------------|

**Solution:**  $P(\text{red}) = 10/40 = 1/4$ .  $P(0) = 4/40 = 1/10$ . With replacement:  $1/4 \times 1/10 = 1/40$ .

**Misconception watch:** 33% correct. B (1/20) was high-frequency wrong — students computed  $P(\text{red AND } 0)$  differently or missed the replacement condition.

|               |                |                                                      |          |
|---------------|----------------|------------------------------------------------------|----------|
| <b>CLUE N</b> | <b>Slot #6</b> | <b>The Grocer's Ledger</b><br>TEKS 7.6(H) · 2025 Q19 | <b>B</b> |
|---------------|----------------|------------------------------------------------------|----------|

**Solution:** Corn (36)  $\div$  Green Beans (24) = 1.5 =  $1\frac{1}{2}$ . So Corn =  $1\frac{1}{2} \times$  Green Beans ✓.

**Misconception watch:** 45% correct. C was the high-frequency wrong answer — students confused the multiplicative comparison with an additive one.

|               |                 |                                                         |             |
|---------------|-----------------|---------------------------------------------------------|-------------|
| <b>CLUE O</b> | <b>Slot #10</b> | <b>The Basketball Records</b><br>TEKS 7.6(H) · 2023 Q35 | <b>LESS</b> |
|---------------|-----------------|---------------------------------------------------------|-------------|

**Solution:** Frequency 116–125 = 17 games < Frequency 106–115 = 21 games. So 116–125 is LESS LIKELY THAN 106–115.

**Misconception watch:** 55% earned full credit on the original; 17% partial — students typically nailed one comparison but missed the other.